

RAILWAY RESERVATION SYSTEM

PHASE 1

REQUIRMENTS:

“Railway Reservation System with Booking and Cancellation Features”, we can organize the requirements into **Functional, Non-Functional, User, Domain, and System Requirements**.

FUNCTIONAL REQUIREMENTS:

- 1.The system shall allow users to register an account.
- 2.The system shall allow registered users to log in and log out.
- 3.The system shall allow users to search for trains based on source, destination, and journey date.
- 4.The system shall display available trains matching the search criteria.
- 5.The system shall display train number, train name, source, destination, departure time, arrival time, and fare.
- 6.The system shall allow users to check seat availability.
- 7.The system shall allow users to select the required class/category of travel.
- 8.The system shall allow users to enter passenger details.
- 9.The system shall calculate the total ticket fare.
- 10.The system shall allow users to book available seats.
- 11.The system shall generate a unique PNR/booking ID after successful booking.
- 12.The system shall display booking confirmation and ticket details.
- 13.The system shall allow users to view their previous and upcoming bookings.
- 14.The system shall allow users to cancel a booked ticket.
- 15.The system shall update seat availability after cancellation.
- 16.The system shall calculate the applicable cancellation/refund amount.
- 17.The system shall display booking/cancellation status to the user.
- 18.The administrator shall be able to add new trains.
- 19.The administrator shall be able to modify train details.
- 20.The administrator shall be able to remove trains from the system.
- 21.The administrator shall be able to manage train routes and schedules.
- 22.The administrator shall be able to view passenger bookings.

- 23.The administrator shall be able to manage seat availability.
- 24.The system shall maintain booking and cancellation records.

Non-Functional Requirements

- 1.The system should provide search results within a few seconds under normal conditions.
- 2 The system should be available whenever users need to make reservations.
- 3 The system should protect user passwords and personal information.
- 4 Only authorized users should be allowed to access their accounts.
- 5 The system should reliably maintain booking and cancellation information.
- 6 The system interface should be simple and easy to use.
- 7 The system should support an increasing number of users and bookings.
- 8 The software should be modular and easy to maintain.
- 9 The system should prevent double booking of the same seat.
- 10 The system should recover from failures without losing confirmed booking data.
- 11 The system should be compatible with commonly used web browsers.
- 12 Passenger information should only be accessible to authorized users and administrators.

3. User Requirements

Sl. No.	User Requirement
1	The passenger should be able to create an account.
2	The passenger should be able to log in securely.
3	The passenger should be able to search for trains between two stations.

4	The passenger should be able to select a journey date.
5	The passenger should be able to view available seats.
6	The passenger should be able to select a preferred travel class.
7	The passenger should be able to enter passenger information.
8	The passenger should be able to book a ticket.
9	The passenger should receive a booking confirmation/PNR.
10	The passenger should be able to view booked tickets.
11	The passenger should be able to cancel a ticket.
12	The passenger should be able to view cancellation/refund status.

Administrator

Sl. No.	User Requirement
13	The administrator should be able to securely log in.
14	The administrator should be able to add and manage train information.
15	The administrator should be able to manage routes and schedules.
16	The administrator should be able to monitor seat availability.
17	The administrator should be able to view passenger reservations.
18	The administrator should be able to view cancellation records.

4. Domain Requirements

Sl. No.	Domain Requirement
1	Every train shall have a unique train number.
2	Every station shall have a unique station identifier/name.
3	A train shall have a defined route consisting of one or more stations.

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| 4 | A train shall have scheduled departure and arrival times. |
| 5 | A seat cannot be assigned to more than one passenger for the same train journey and date. |
| 6 | Every ticket shall have a unique PNR/booking number. |
| 7 | A booking shall contain passenger details and journey information. |
| 8 | A ticket can only be booked when seats are available. |
| 9 | Cancellation shall follow the defined railway cancellation/refund policy. |
| 10 | Once a ticket is cancelled, the corresponding seat should become available. |
| 11 | Fare calculation shall follow the fare rules defined by the system. |
| 12 | A passenger cannot cancel a booking that does not exist. |
| 13 | The system must maintain records of confirmed and cancelled reservations. |

System Requirements

Sl. No.	Hardware Requirement
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| 1 | Processor: Dual-core processor or better. |
| 2 | RAM: Minimum 4 GB. |
| 3 | Storage: Minimum 10 GB free space. |
| 4 | Network connection for web-based deployment. |

Sl. No.	Software Requirement
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| 1 | Operating System: Windows, macOS, or Linux. |
| 2 | Frontend: HTML, CSS, JavaScript or React. |
| 3 | Backend: Java, Python, Node.js, or PHP. |

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| 4 | Database: MySQL. |
| 5 | Web Server: Apache or Node.js server. |
| 6 | Web Browser: Chrome, Firefox, Edge, or Safari. |
| 7 | Development Environment: VS Code, IntelliJ, or Eclipse. |

System-Level Requirements

Sl. No.	System Requirement
1	The system shall authenticate registered users using valid credentials.
2	The system shall prevent unauthorized access to user and administrator functions.
3	The system shall accept source, destination, and journey date as search inputs.
4	The system shall display trains matching the search criteria.
5	The system shall display available seats for each train.
6	The system shall allow users to select an available train and class.
7	The system shall collect passenger details.
8	The system shall verify seat availability before confirming a booking.
9	The system shall generate a unique PNR after successful booking.
10	The system shall update available seats after booking.
11	The system shall allow users to select an existing booking for cancellation.
12	The system shall verify that the booking belongs to the logged-in user.
13	The system shall calculate the applicable refund.
14	The system shall update the booking status to "Cancelled".
15	The system shall update seat availability after cancellation.
16	The system shall provide an administrator interface.

- 17 The administrator shall be able to add, update, and delete train information.
- 18 The administrator shall be able to manage schedules and routes.
- 19 The administrator shall be able to view booking and cancellation records.

Actors and Goals

	Actor	Goals
1	Passenger / User	Register an account, log in, search trains, check seat availability, book tickets, view bookings, cancel tickets, and check refund status.
2	Administrator	Manage trains, stations, routes, schedules, fares, seats, bookings, cancellations, and passenger information.
3	Payment System	Process ticket payments, confirm successful transactions, and provide payment status.
4	Railway Reservation System	Maintain train, passenger, seat, booking, cancellation, and payment information and ensure correct reservation processing.

FLOW DIAGRAM

